Adaptive Estimation for Weakly Dependent Functional Time Series

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Motivation

We study the local regularity of weakly dependent functional time series (FTS) where the sample paths are observed with error at possibly random, design points. Non-asymptotic concentration bounds of the regularity estimators are derived. As an application, we build nonparametric mean and autocovariance functions estimators that adapt to the regularity of the sample paths and the design which can be sparse or dense. We also derive the asymptotic normality of the adaptive mean function estimator. An extensive simulation study and a real data application illustrate the good performance of the new estimators.

Weak dependency

Let $\boldsymbol{X} = (X_n)_{n\in\mathbb{Z}}$ be a stationary FTS, with continuous paths, defined on the interval $I = [0,1]$.

We reconsider $\mathbb{L}^p_{\mathcal{H}} - m-$*approximability* introduced by by Hörmann and Kokoszka (2010) in context of $\mathcal{C}$-valued random processes. *"The general idea is to approximate X_n by a m-dependent sequence and to obtain the limiting behaviour of the original process from that of its coupled one."*

Example. *FAR(1) is $\mathbb{L}^p_{\mathcal{C}} - m$-approximable:*

$$X_n(t) = \int_0^1 \beta(t,s)X_{n-1}(s)ds + \varepsilon_n(t)$$

$\{\varepsilon_n\}_{n\in\mathbb{Z}}$ are i.i.d. fBm with Hurst exponent H_ε.

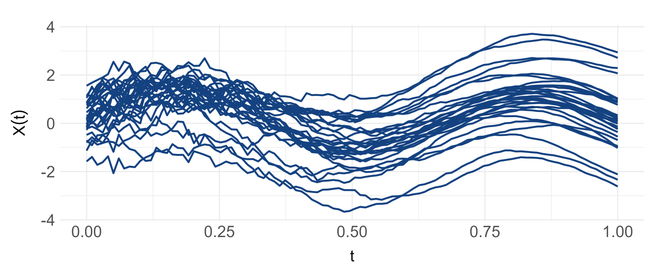
The local regularity parameters

The process X, with non differentiable paths, admits a ***local regularity*** at $t \in I$, with

- local **exponent** $H_t \in (0,1)$,
- and local **Hölder constant** $L_t > 0$, if

$$\mathbb{E}\left[(X(u) - X(v))^2\right] \approx L_t^2|u-v|^{2H_t},$$

$\forall u, v \in [t-\Delta/2, t+\Delta/2]$ for some small $\Delta > 0$.



Varying regularity across the domain.

Data observation Framework

For $n = 1\dots N$, the trajectory X_n is measured with error at discretely, randomly sampled, domain points:

$$Y_{n,k} = X_n(T_{n,k}) + \varepsilon_{n,k}, \quad 1 \le k \le M_n,$$

- $M_1, \dots, M_N \overset{i.i.d.}{\sim} M$ with expectation λ,
- the observation times $T_{n,k} \sim T$ are independent,
- $\varepsilon_{n,k} \sim \epsilon$ are independent centered errors,
- $\mathbf{X}$, M and T are mutually independent.

The local regularity estimators

With $\widetilde{X}_n$ a nonparametric estimator of X_n, let u, v close to t,

$$\widehat{\theta}(u,v) = \frac{1}{N}\sum_{n=1}^{N}\left\{\widetilde{X}_n(v) - \widetilde{X}_n(u)\right\}^2.$$

Let $t_1 = t - \Delta/2$, $t_3 = t + \Delta/2$. The estimators of H_t and L_t^2 are

$$\widehat{H}_t = \frac{\log(\widehat{\theta}(t_1,t_3)) - \log(\widehat{\theta}(t_1,t))}{2\log(2)} \quad \text{and} \quad \frac{\widehat{\theta}(t_1,t_3)}{\Delta^{2\widehat{H}_t}}.$$

Concentration bounds

Let $\{X_n\}$ be $\mathbb{L}^4_{\mathcal{C}} - m$-approximable. Assume that the $\mathbb{L}^2$-risk of smoothing is suitably bounded. Then, for any $\lambda \ge \lambda_0$, for some λ_0, and for $\Delta > 0$ and $\varphi > 0$ depending on λ, we have

$$\mathbb{P}\left(|\widehat{H}_t - H_t| > \varphi\right) \le \frac{4\mathfrak{f}_1}{N\varphi^2\Delta^{4H_t}} + 4\mathfrak{b}\exp\left(-\mathfrak{f}_2 N\varphi^2\Delta^{4H_t}\right),$$

where $\mathfrak{b} > 0$ is a constant and $\mathfrak{f}_1, \mathfrak{f}_2$ are also constants depending on the dependence measure.

NB: a similar result is obtained for $\widehat{L_t^2}$.

Mean function estimation

Let $\mu(t) = \mathbb{E}(X_n(t))$.

- **The objective:** estimation of $\mu(t)$ by adaptive smoothing of X_n.
- The proposed estimator is $\widehat{\mu}_N(t;h^*_\mu)$, with

$$\widehat{\mu}_N(t;h) = \sum_{n=1}^{N}\frac{\pi_n(t;h)}{P_N(t;h)}\widehat{X}_n(t;h), \qquad \text{where}$$

  - $\pi_n(t;h) = 1$ if $\#\{T_{n,i} \in [t-h, t+h]\} \ge 1$ and 0 else,
  - $P_N(t;h) = \sum_{n=1}^{N}\pi_n(t;h)$,
  - $\widehat{X}_n(t;h)$ is NW estimator with bandwidth h.
- Derive a sharp bound of the quadratic risk :

$$\mathbb{E}_{M,T}\left[(\widehat{\mu}_N(t;h) - \mu(t))^2\right] \le 2R_\mu(t;h),$$

$$R_\mu(t;h) = L_t^2 h^{2H_t}\mathbb{B}(t;h,2H_t) + \sigma^2(t)\mathbb{V}_\mu(t;h) + \frac{\mathbb{D}_\mu(t;h)}{P_N(t;h)},$$

and define $\quad h^*_\mu \in \arg\min_{h\in\mathcal{H}_N} \widehat{R}_\mu(t;h).$

Convergence rate of the mean

Let $t \in I$. Under some assumptions we have

$$\widehat{R}_\mu(t;h) = \mathcal{O}_{\mathbb{P}}\left\{h^{2H_t} + (N\lambda h)^{-1} + N^{-1}\right\},$$

$$h^*_\mu = \mathcal{O}_{\mathbb{P}}\left\{(N\lambda)^{-\frac{1}{1+2H_t}}\right\},$$

and the estimator $\widehat{\mu}_N(t;h^*_\mu)$ satisfies

$$\widehat{\mu}^*_N(t;h^*_\mu) - \mu(t) = \mathcal{O}_{\mathbb{P}}\left\{(N\lambda)^{-\frac{H_t}{1+2H_t}} + N^{-1/2}\right\}.$$

(Auto)covariance estimator

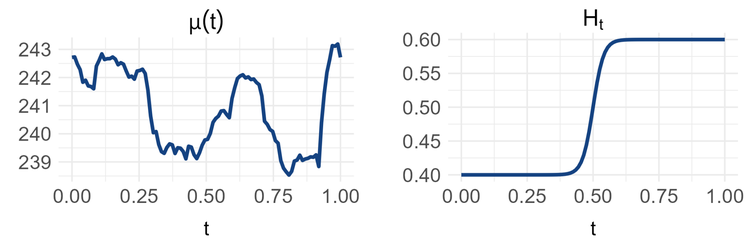
Similar idea to mean estimation. Scan more details.

Simulation

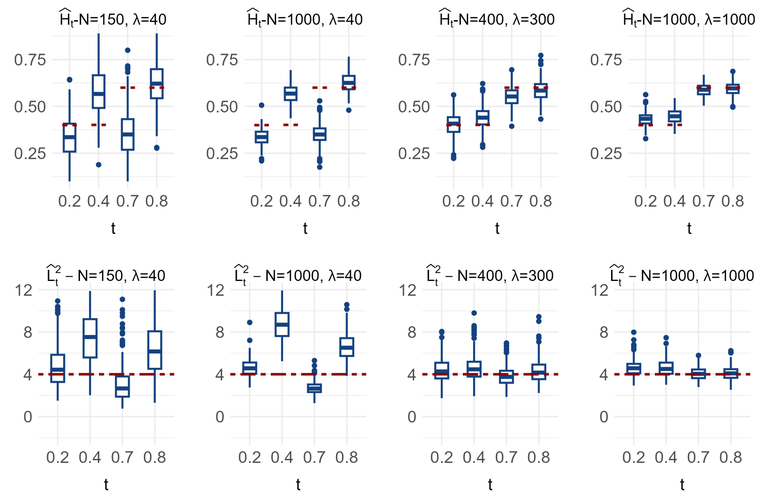
We simulate a FAR(1) where $\{\varepsilon_n\}$ are i.i.d. *multifractional Brownian motion* (see [5]) paths with :

- $H_t \in [0.4, 0.6]$ a logistic function and $L_t^2 = 4$,
- $\beta(s,t) = 1.13\exp(-(s+2t)^2)$.

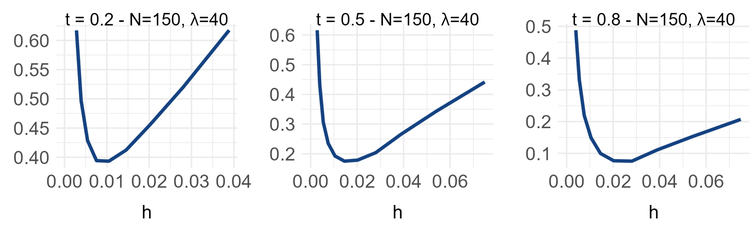
Left. True mean function. **Right.** Hurst function.



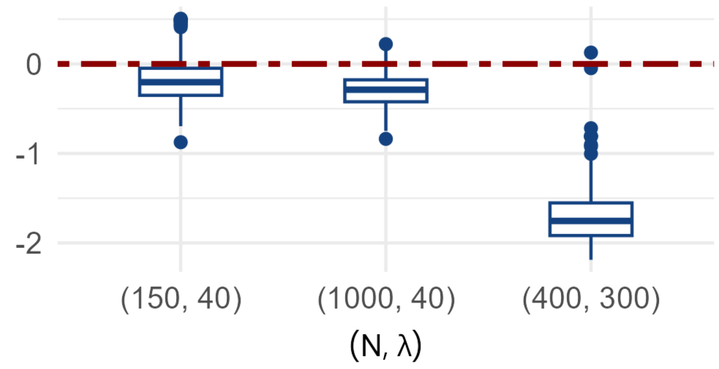
Local regularity estimates over 400 replications.



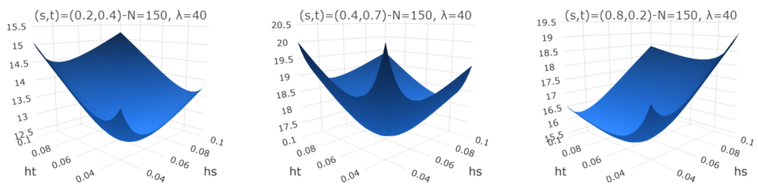
Average of $\widehat{R}_\mu(t;h)$ over 400 replications.



Mean function estimates comparison.
log of the ISE ratio of our method / [4]



Risk function – lag-1 autocov estimator
Average over 400 replications.



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